

Changelog: Job Churn Among Foreign-Born Workers in Sweden

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Project overview

Question: Do foreign-born workers in Sweden change jobs more often than Swedish-born workers, and has that changed over time?

- **Course:** Google Data Analytics Certificate, capstone (people analytics case study)
- **SQL tool:** SQLite, using DB Browser for SQLite
- **Database file:** sweden.churn.db (backup of first attempt: sweden_churn_OLD_backup.db)
- **Visualization:** Tableau Public
- **Churn measure:** share of employees who started their current job within the past 12 months ("newly employed")

Data sources

Both datasets come from the EU Labour Force Survey, published by Eurostat. Reuse is allowed with attribution: "Source: Eurostat, dataset [code]."

Table name	Eurostat code	What it measures	Years	Used in analysis	Downloaded
newly_empl	lfsa_enewasc	Share of employed persons in their current job 12 months or less, by country of birth	1995–2025 (birth groups from 1997)	Yes (core measure)	2026-09-25
migration_job_tenure	lfsa_eegatm	Employees by migration status and contract type (thousands)	2021–2025	Kept, not yet used	2026-09-25

Metadata reviewed: Eurostat reference metadata for Employment and unemployment (LFS), last updated 13 March 2024; Statistics Sweden national reference metadata, last updated 15 May 2025.

Understanding the data structure

The files use the SDMX (Statistical Data and Metadata eXchange) format, which is a standard used by Eurostat, the OECD, the IMF and central banks. Each row is one observation: one value for one combination of country, year and group.

- **OBS_VALUE:** the number being analyzed (here, a percentage)
- **OBS_FLAG:** a data quality note on that value; empty means nothing unusual
- **TIME_PERIOD:** the year the value refers to
- **Dimension columns** (geo, age, c_birth, wstatus and so on): which group the value belongs to

Flag Meaning

- b Break in time series (survey method changed)
- u Low reliability, usually a small sample
- p Provisional, may be revised
- e Estimated
- c Confidential, value hidden
- d Definition differs from the standard

A colon (:) in the value column means data not available.

Processing log

Newest entries first.

Date	Step	What was done
2026-09-28	Saved visualization	Workbook saved to Tableau Public profile.
2026-09-28	Built charts	Sheet 1: line chart of Swedish-born vs. foreign-born share by year. Sheet 2: bar chart of the gap. Dotted reference lines at 2001, 2005, 2018 and 2021; titles and captions added.

2026-09-28	Exported data	Exported churn_by_year to churn_by_year.csv for Tableau.
2026-09-28	Added gap column	ALTER TABLE added gap; UPDATE set gap = foreign-born minus Swedish-born, rounded to 1 decimal.
2026-09-28	Reshaped by year	Created churn_by_year (02_churn_by_year.sql): one row per year with swedish_born, foreign_born and flag columns, using MAX(CASE WHEN ...) with GROUP BY year. 29 rows (1997–2025).
2026-09-28	Created clean table	Created churn_sweden (01_clean_churn.sql) from newly_empl: Sweden, Employees, age 20–64, Reporting country and Foreign country. Converted OBS_VALUE to REAL; renamed columns to year, birth_group, newly_employed_pct, flag. 58 rows (29 years × 2 groups), as expected.
2026-09-28	Restarted with simpler scope	Backed up the database, then deleted all tables except newly_empl and migration_job_tenure. Old SQL files moved aside.
2026-09-28	Reviewed metadata	Read Eurostat and Statistics Sweden metadata to understand series breaks and coverage (see sections below).
2026-09-25	Imported raw data	Imported Eurostat CSVs into sweden.churn.db. Raw tables kept unchanged; all cleaning done in new tables.

Project files

File or table	Type	Contents
newly_empl	Raw table	Eurostat Ifsa_ewasc, as downloaded
migration_job_tenure	Raw table	Eurostat Ifsa_eegatm, as downloaded
churn_sweden	Clean table	Sweden, employees 20–64, two birth groups, 58 rows
churn_by_year	Analysis table	One row per year with both groups, flag and gap, 29 rows
01_clean_churn.sql	SQL file	Creates churn_sweden
02_churn_by_year.sql	SQL file	Creates churn_by_year and adds the gap column

churn_by_year.csv	Export	Data used in Tableau
Tableau Public workbook	Visualization	Line chart and gap bar chart

Decisions and assumptions

- **Scope:** narrowed to one core measure (newly employed share) after a first attempt with six datasets became too broad for a capstone.
- **Area:** Sweden only; the EU comparison was dropped to keep the project focused.
- **Employees only:** self-employed people and family workers excluded, since they don't change employers in the HR sense.
- **Age 20–64:** standard working-age group, available in all datasets.
- **Groups:** "Reporting country" = Swedish-born; "Foreign country" = foreign-born. EU and non-EU subgroups, "No response" and "Total" excluded.
- **Years:** full series, 1997–2025 (the first year with a country-of-birth breakdown).
- **Units:** percentages, so groups of different sizes can be compared.

Survey method changes (series breaks)

Eurostat flags four break years ("b") in Sweden's series. Values just before and after a break are not fully comparable, so each break is marked with a dotted line on the charts.

Year	Likely cause	Source
2001	Classification changes in Sweden's survey (sector classification INSEKT 2000)	ILO profile of Sweden's Labour Force Survey
2005	New EU-adapted Labour Force Survey introduced	ILO profile of Sweden's Labour Force Survey
2018	Cause not documented in the sources reviewed	Eurostat flag only
2021	Survey adapted to EU regulation 2019/1700 (new labour status definitions)	Statistics Sweden, labour market ; Eurostat metadata

Comparable segments: 1997–2000, 2001–2004, 2005–2017 (longest), 2018–2020 and 2021–2025.

Known limitations

- **What "newly employed" captures:** it counts both people who switched jobs and people who entered work for the first time or after a gap. For recent immigrants, many "new" jobs are first jobs in Sweden, so the measure reflects labor market entry as well as turnover which I want to keep into consideration.
- **Survey method changes:** in 2001, 2005, 2018 and 2021, part of any jump may come from the method rather than real change.
- **Definition of foreign-born:** includes everyone born abroad, whether they arrived as a young child or as an adult looking for work. Public tables don't show age at arrival.
- **Residence permits:** people without a residence permit (for example, asylum seekers awaiting a decision) are not in Sweden's population register, so they aren't in the survey.
- **Aggregate data only:** the data shows group averages, not individual employees, so it can't explain why a given person changed jobs.
- **Sample survey:** about 0.3% of the population is sampled each quarter, and Statistics Sweden reports declining response rates.

Findings so far

Foreign-born employees in Sweden were more likely to be new in their job than Swedish-born employees in every year from 1997 to 2025. In 2025 the shares were 19.4% and 14.1%, about 1.4 times as high.

- **Persistent gap:** the difference ranged from 3.9 percentage points (2002) to 9.4 (2018).
- **Moves with the economy:** both groups dropped in 2009 (financial crisis) and 2020 (COVID), and both fell from 2022 to 2025.
- **Widened, then narrowed:** within the comparable 2005–2017 segment, the gap rose from about 5.5 to 7 points. It was 8–9 points in 2018–2020 and narrowed to about 5 by 2023–2025.
- **Possible explanation, not proven:** the late-2010s rise may partly reflect people who arrived in 2015–2016 entering jobs. 2018 is also a break year, so some of that jump may come from the survey change.